Question:

A food caterer was told to prepare gourmet hot dogs for 36 guests. While most people would only eat one hotdog, he prepared enough for half of the guests to be able to have two hotdogs. However, 40 guests showed up, and everyone wanted a second hotdog. How many guests did not get a second hotdog?

Reasoning + Answer:

<think>The caterer was told to prepare gourmet hotdogs for 36 guests, so he prepared enough for half of the guests to be able to have two hotdogs, which means: \[\frac{36}{2} = 18 \text{hotdogs for the first half of the guests} \] Since 40 guests showed up, and everyone wanted a second hotdog, the number of hotdogs needed is: \[40 \text{hotdogs} \] The number of hotdogs the caterer prepared is enough for the first 18 guests to have two hotdogs each, which totals: \[18 \times 2 = 36 \text{hotdogs} \] Since only 36 hotdogs were prepared and 40 guests showed up, the number of guests who did not get a second hotdog is: \[40 - 18 = 22 \text{guests} \]</think><answer>22</answer>